



Fold and Go Knee Scooter for Increased Single Leg Mobility

Fold & Go

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Abstract

- Over 23,000 reported ankle sprains occur every day
- Over 6.8 million Americans are currently using a mobility device
 - Over 40% of them are unable to perform desired activities
 - Complaints include versatility, portability and weight
- Mobility device market is projected to grow to 8 billion USD by 2018
- *Solution: Develop a portable and light weight knee scooter to provide patients with an alternative to current single leg mobility devices that are on the market today.*

Current Solutions

Crutches



- Slow
- Limited terrain
- Uncomfortable
- Dangerous

Prosthetics



- Slow
- Unstable
- Difficult to use
- Expensive

Knee Scooter



- Heavy (~25lbs)
- Large (~ 8.9ft³)
- Expensive
- Difficult to fold

Design Concept

Key Design Considerations and Targets:

Lightweight construction

- Constructed of 6061-T6 Aluminum
- Weight Target of < 19lbs

Collapsible architecture

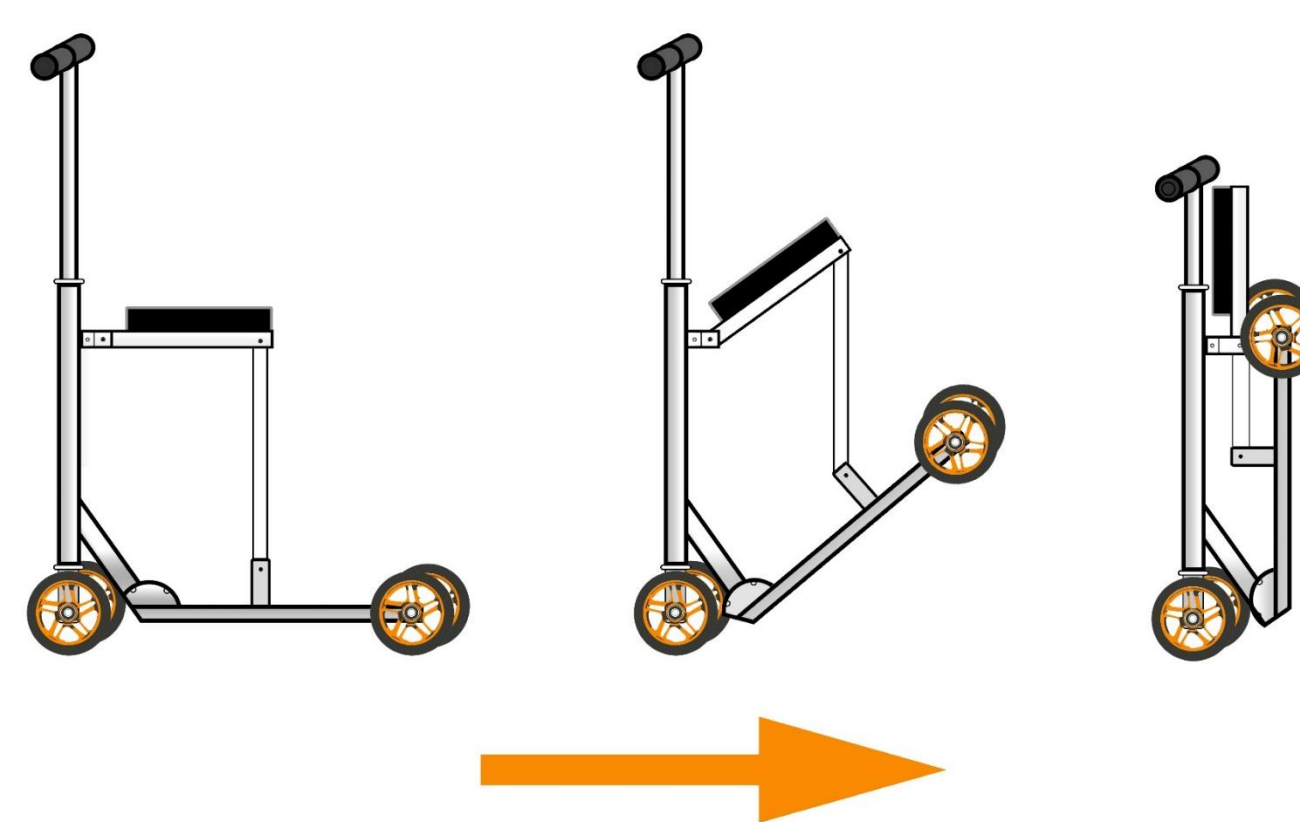
- Three pivot points allows the device to fold flat when not in use
- Locking mechanism allows for one step operation

Ability to support a 250lb user

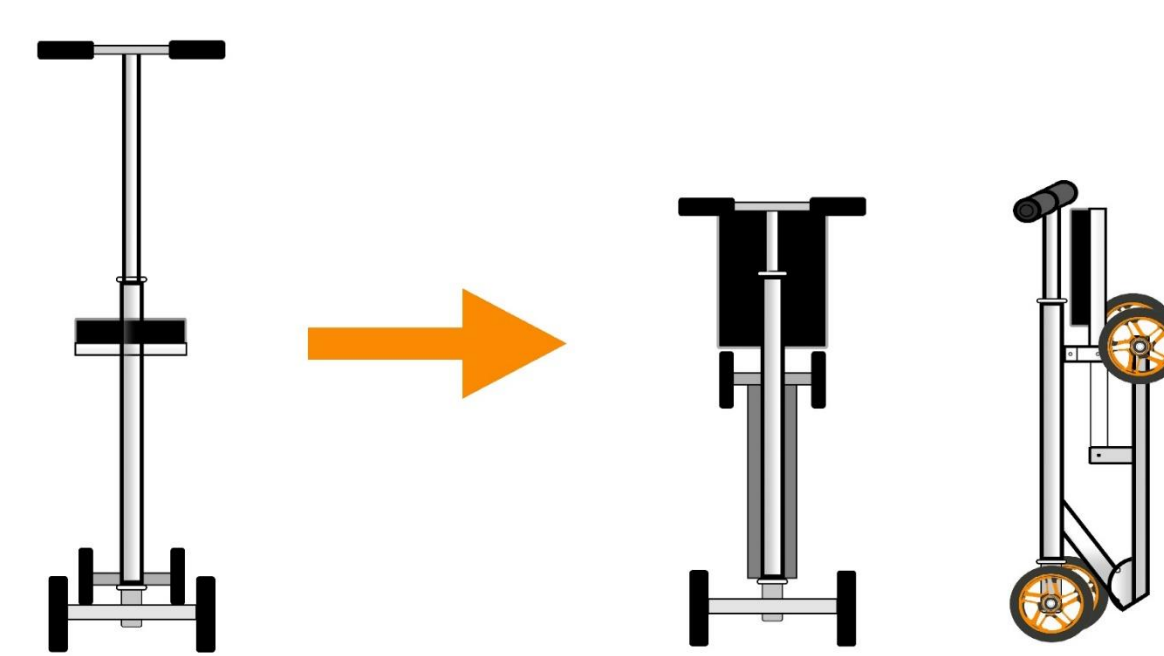
- Testing done at 500lbs to account for safety factor and misuse

Inexpensive to the end user

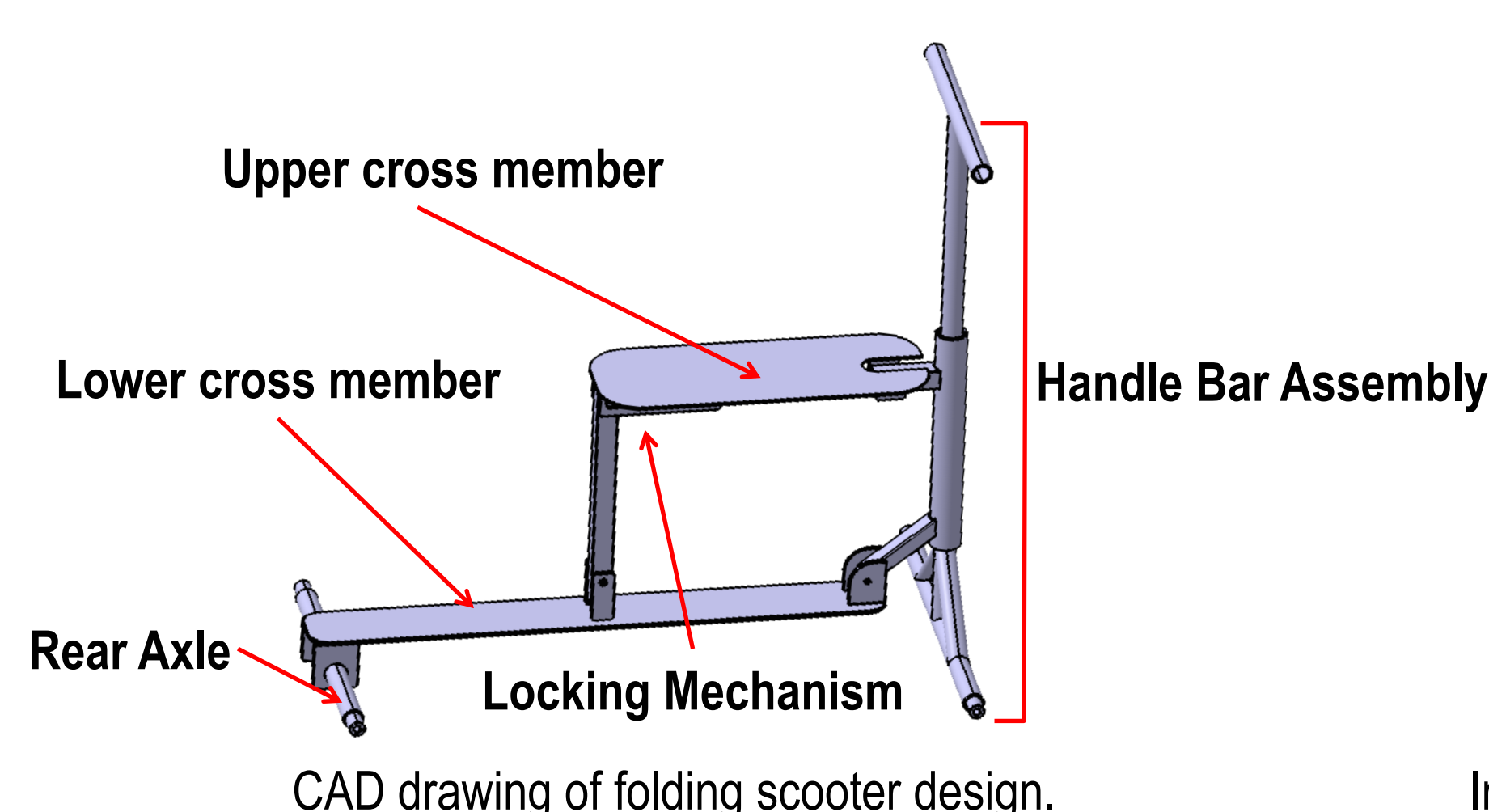
- Cost target of < \$200.00



Side profile view rendering of folding concept.



Front view rendering of folding concept.



CAD drawing of folding scooter design.



Initial prototype constructed of wood and PVC.

Testing Methods

Lower Platform and Pivot Bolt Materials Testing

- Insert and level lower cross member in test fixture
- Apply compression load until a 500lbf is achieved
 - Load applied to both pivot bolt and lower member
- Tests performed on 3 lower member designs
- Record and compare force v displacement data

Lower Platform Testing Setup



Flat lower platform
mass = 0.49kg

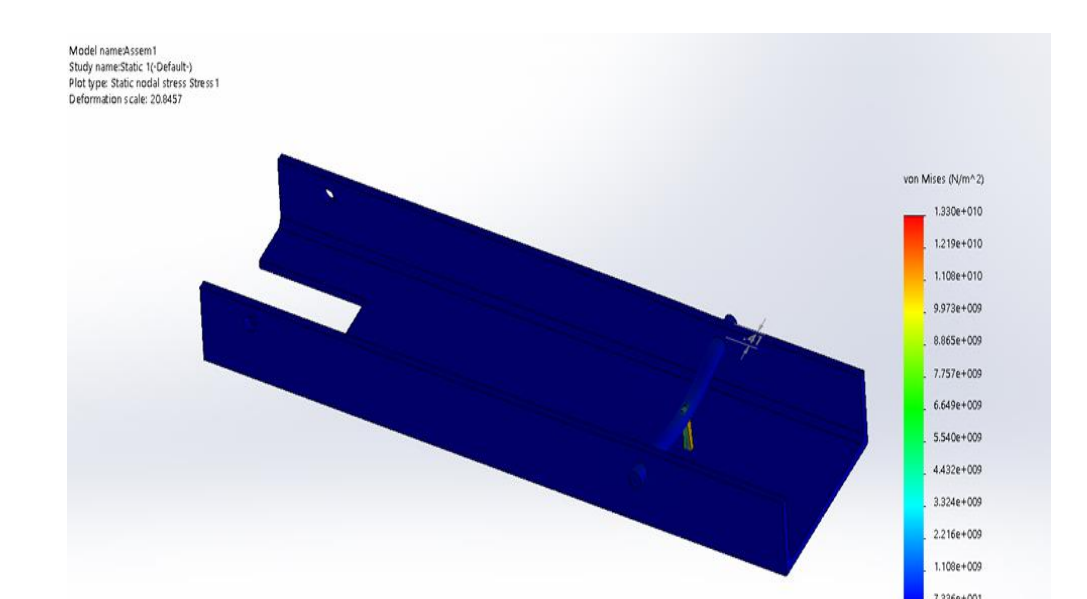
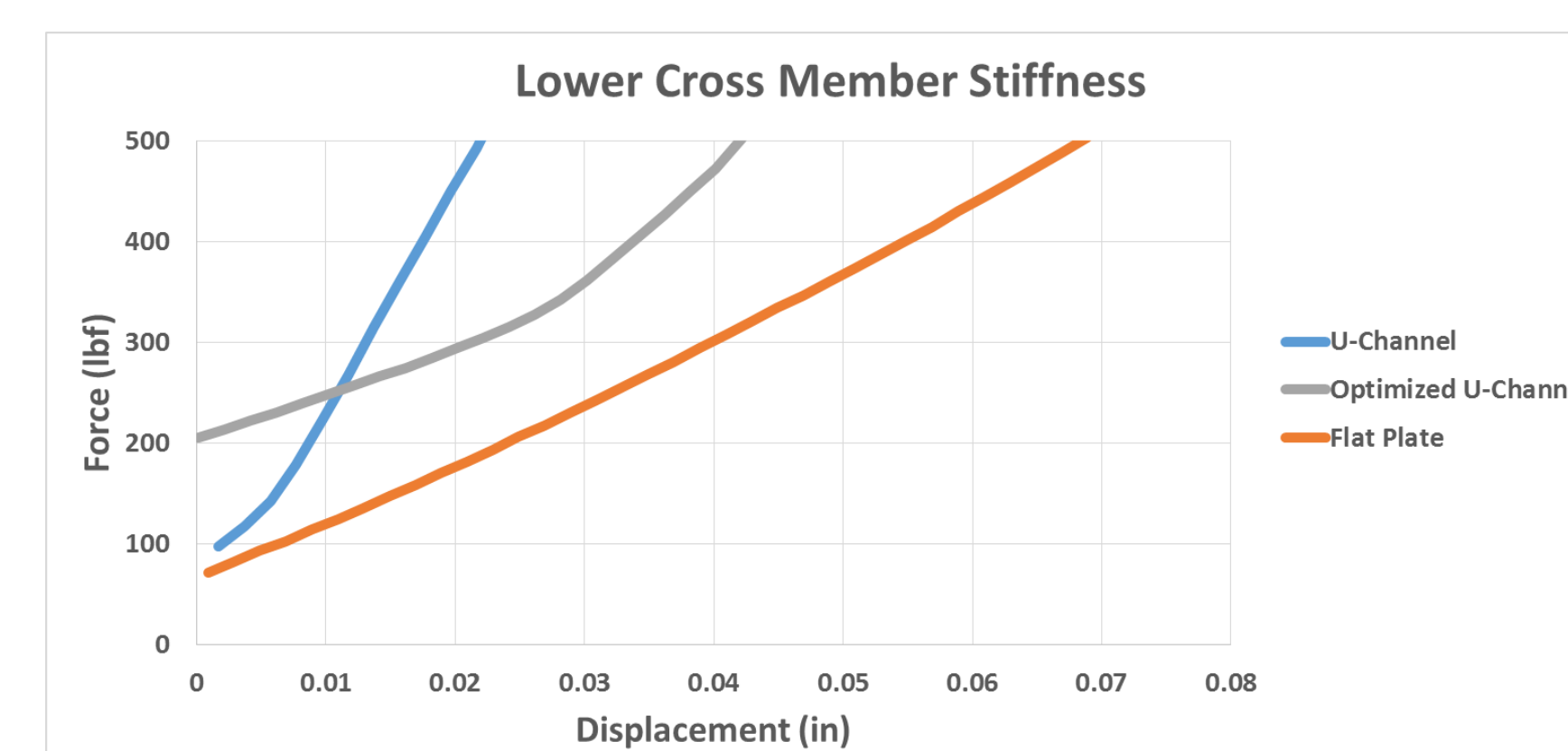
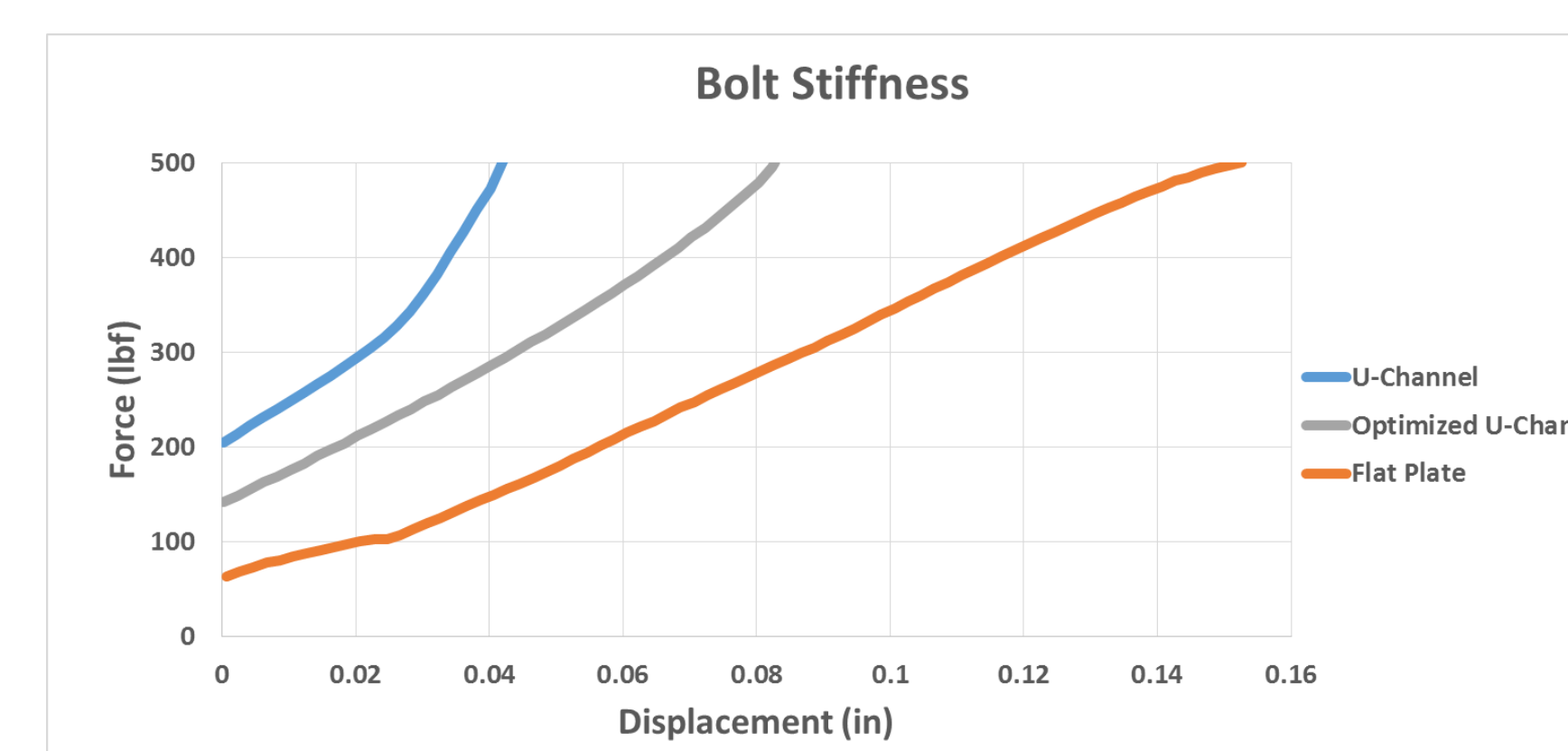


U-channel lower platform
mass = .79kg

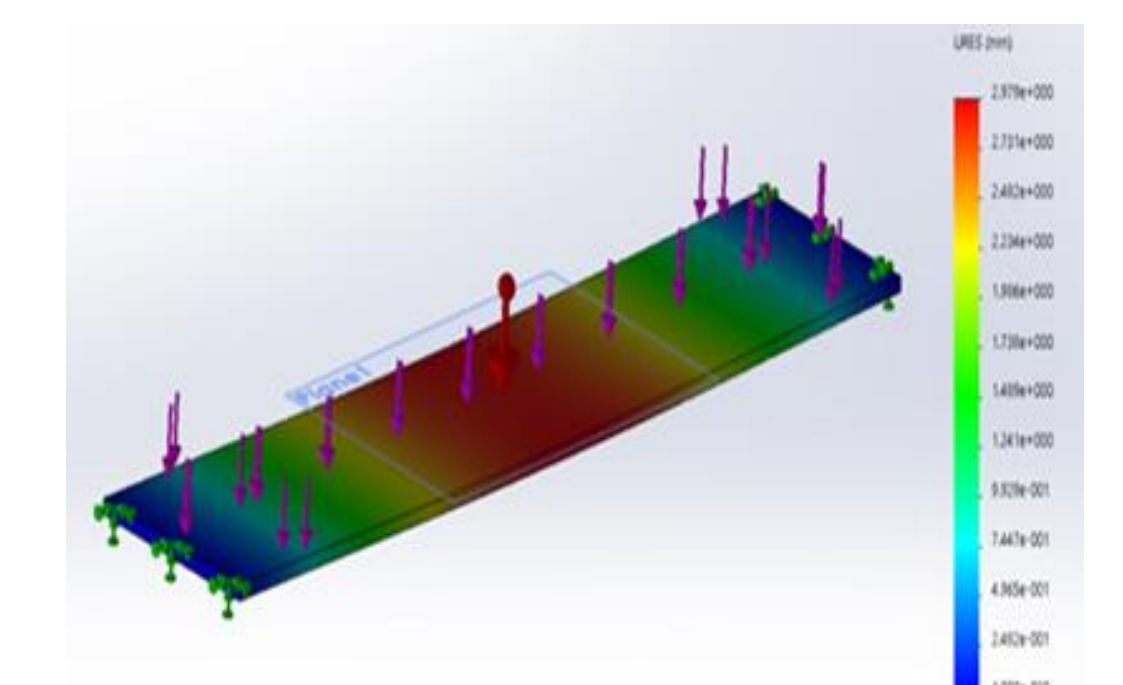


Optimized U-channel platform
mass = .60kg

Testing Results



FEA results of lower cross member pivot bolt with U-channel design.



FEA results of flat lower cross member without pivot bolt.

Prototype Results



Total weight of scooter system (including brakes, brake lever and hand grips) = 15lbs
Package space of scooter system = 4.6ft³

- Optimized u-channel design showed minimal deformation
- One step folding design achieved
- Weight target of <19lbs was achieved
- Package space reduced by 45%
- Added brake system

Future Work

- Investigate a more user friendly spring loaded locking mechanism
- Investigate the use of thinner walled aluminum material
- Conduct subjective reviews to improve the design even further
- Apply for a patent

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